PPE:
PERSONAL PROTECTIVE EQUIPMENT

INTRODUCTION

Depending on the work environment, you could be exposed to various hazards if the proper precautions are not taken. When precautions such as elimination, substitution, isolation, administrative controls, and/or engineering controls do not protect employees from hazards, personal protective equipment (PPE) is used as the last line of defense. In this lesson, we will cover common hazards that may be found in the working environment, the different types of personal protective equipment, when personal protective equipment should be worn, the inspection of personal protective equipment, and the maintenance of personal protective equipment.

HAZARDS

Note: The list of hazards mentioned in this lesson is not all encompassing and more may exist in your work environment. If you have questions about hazards associated with your work, please speak with your supervisor.

Common hazards that are found in many work environments include:

- Eye irritation or damage
- Skin irritation or damage
- Respiratory irritation or damage
- Cuts, lacerations, or punctures
- Head injuries
- Hearing loss or damage
- Burns
- Electrical shock or electrocution
- Suffocation
- Slips, trips, or falls
- Frostbite
- Radiation exposure
- Exposure to chemicals or solvents
- Illnesses related to heat exposure
- Bruising
- Fractures

TYPES OF PERSONAL PROTECTIVE EQUIPMENT

Due to the variety of hazards that may be present in the workplace, there are many different types of personal protective equipment that you may use to help minimize the chances of an accident occurring. These types include:

- Head protection which may consist of:
  - Hard hat
  - Bump cap
  - Helmets
- Eye protection which may consist of:
  - Safety glasses (prescription or non-prescription)

  - Goggles
  - Laser safety glasses
- Face protection which may consist of:
  - Face shield
  - Welding shield
- Hand protection which may consist of:
  - Disposable gloves such as vinyl, nitrile, or latex
  - Chemical resistant gloves
  - Insulated gloves
  - Cut/puncture-resistant gloves
  - Electrically insulated gloves
- Body protection which may consist of:
  - Long sleeved clothing
  - Close-fitting clothing
  - Reflective vests or jackets
  - Chemical/Hazmat suits
  - Coveralls
  - Aprons
  - Chaps
  - Waders
  - Leg guards
- Respiratory protection which may consist of:
  - Nuisance dust mask
    Note: While nuisance dust masks can help protect against respiratory irritation from regular dust, these types of masks are not considered to be respirators and should not be worn in situations where a respirator is needed.
  - Filtering facepiece respirator (commonly known as N95s)
  - Half-piece respirator
  - Full face respirator
  - Powered air-purifying respirator (commonly known as PAPRs)
  - Supplied-air respirator
  - Self-contained breathing apparatus (SCBAs)
  - Combination respirator
- Foot protection which may consist of:
  - Slip-resistant shoes
  - Shoe covers
  - Steel or composite-toed shoes
  - Waterproof shoes
  - Puncture resistant shoes
  - Chemical resistant shoes
  - Thermally insulated shoes
  - Dielectric electric overshoes
  - Metatarsal guards
  - Static dissipative shoes
  - Electrical hazard-rated shoes
- Fall protection which may consist of:
  - A personal fall-resistant system
  - A personal fall arrest system

Note: For more in-depth information on specified categories of personal protective equipment mentioned in this lesson, please refer to our PPE series of safety lessons.

WHEN PERSONAL PROTECTIVE EQUIPMENT SHOULD BE WORN

To help identify when personal protective equipment should be worn, a job hazard or task hazard analysis should be conducted to identify hazards that exist in the workplace. In addition to the analysis, manufacturer-provided safety data sheets (SDS) and operator/owner manuals should also be consulted for information related to recommended or required PPE.

Where an identified hazard exists and cannot be eliminated, appropriate PPE should be worn to help minimize the chances of an accident occurring.

INSPECTION OF PERSONAL PROTECTIVE EQUIPMENT

Where PPE is recommended or required for the work, you should inspect all provided PPE for wear or damage before you put it on. Inspections should include looking for:

- Cracks
- Tears
- Fraying
- Chips
- Scratches
- Scrapes
- Pulled or missing stitches
- Cuts
- Punctures
- Discoloration
- Holes
- General wear

All PPE items that have wear or damage should be reported to your supervisor and not worn until it can either be repaired or replaced. Do NOT wear damaged PPE.

Additionally, you should make sure that the PPE fits you properly and comfortably. If something is not fitting right, please speak with your supervisor.

MAINTENANCE OF PERSONAL PROTECTIVE EQUIPMENT

To help make sure that provided PPE lasts and does its job of adequately protecting you from workplace hazards, all provided PPE should be cleaned and properly maintained. PPE that you wear should be cleaned both before use and after use and cleaned in accordance with the manufacturer's instructions and your company's policies and procedures.

When you are not wearing PPE, it should be stored in the appropriate areas in accordance with the manufacturer's instructions and your company's policies and procedures.

CONCLUSION

To conclude, when hazards cannot be eliminated in the workplace, personal protective equipment (PPE) is used to minimize the chances of an accident occurring as a last line of defense. PPE should be worn and maintained in accordance with the manufacturer's instructions and your company's policies and procedures. If you have questions regarding PPE at your workplace, please speak with your supervisor.

EXCAVATION: DUST

INTRODUCTION

Dust is part of the construction industry no matter where one is working. Dust such as this is also known as Fugitive dust and can be part of the environment or be created during the construction project. Dust may seem like a minor issue since we encounter it in our day to day lives; however, dust can cause many different health problems if not taken care of properly at a construction or excavation site. Following the safe work practices provided in this lesson will help ensure employee safety.

TYPES OF DUST SUPPRESSION

There are multiple ways in which dust can be contained or suppressed during an operation, including the addition of:

- Vegetation and/or Mulch
- Tackifiers (Resins/Chemical Compounds)
- Misted Water
- Barriers and Fencing
- Applied Polymers

HAZARD

Dust can expose employees to several hazards, including:

- Valley Fever (Southwestern United States)
- Silicosis
- Respiratory problems
- Polluting the air
- Damage to equipment

PERSONAL PROTECTIVE EQUIPMENT (PPE)

Employees should wear the following personal protective equipment:

- Safety glasses
- Comfort mask (optional)
- Respirator (when required)

EXCAVATION: DUST

SAFE WORK PRACTICES

- Read and understand the operator manual for all dust suppression equipment.
- Inspect all dust suppression equipment for any broken or malfunctioning parts.
- Report any found malfunctioning or broken parts to your competent person or supervisor, and do not use that specific piece of equipment.
- Before the shift begins, check the day’s forecast to see what the weather conditions will be.
- Look for any signs of a high wind.
- If high winds have kicked up before the operation, use your dust suppression system to ensure that minimal or no dust will be kicked up during the operation.
- Should high winds occur during the operation, stop the operation and allow for the dust suppression equipment to be used.
- When moving loads of dirt from one place to another using a truck, ensure that the dirt has been covered to help minimize the dust.
- Storage areas of dirt should be covered to help minimize or eliminate dust.
- When using a water suppression system, the soil should only be damp, not soaking.
- Record all dust activity.
- All equipment should be hosed/cleaned off at the end of the day.

CONCLUSION

Employees will find that they will not have to worry about dust as often when they follow the safe work practices.

NOISE-INDUCED HEARING LOSS

INTRODUCTION

Noise-induced hearing loss (NIHL) is a condition that can affect millions of employees in many different industries. This condition is one that can develop over time or it can happen suddenly, and the symptoms or indicators are often ignored because of how subtle the changes can be. Employees usually develop this condition from not wearing the appropriate hearing protection while being exposed to loud noises for extended periods of time. The good news for employees is that there are various types of hearing protection available and that NIHL is preventable.

NOISES THAT CAN CAUSE DAMAGE

NIHL occurs when an employee is exposed to loud noises while working. The level of sound can vary from work area to work area. Some items or situations that may raise risk of an employee developing NIHL include:

- Machinery
- Power tools
- Loud music
- Explosions
- Fire arms
- Power weed cutters
- Blowers

HOW SOUND IS MEASURED

Sound is measured in decibels. When employees are exposed to sounds above 90 decibels, they have a higher risk of developing NIHL over the course of their career. Some examples of environments or pieces of equipment that exceed 90 decibels include:

- Loud bass in cars – 120 decibels
- Snowmobiles – 120 decibels
- Rock concerts and firecrackers – 140 decibels
- Chain saws – 110 decibels
- Wood shops – 100 decibels

Employees who work in environments that range from 90 decibels and lower have a lower risk of developing NIHL. Some examples of environments or pieces of equipment that do not exceed 90 decibels include:

- Lawn mowers – 90 decibels
- Motorcycles – 90 decibels
- City traffic – 80 decibels
- Normal conversations – 60 decibels
- Refrigerator humming – 40 decibels

INDICATORS OF NIHL

Employees may be exposed to decibels that are above 90 and not even know it. The following are some indicators that employees are in an environment that may require hearing protection:

- If it is necessary to speak in a very loud voice or shout.
- If you hear noises or ringing in the ears at the end of the work day.
- If speech or music sounds muffled to you after leaving work but sounds fairly clear when you return to work the following morning.

NOISE-INDUCED HEARING LOSS

TYPES OF HEARING PROTECTION

Employees have a variety of hearing protection to choose from. Environment, machinery/equipment, and fit are all factors that should be taken into consideration when choosing the appropriate hearing protection for the job. Some examples of appropriate hearing protection include:

- **Active Noise Cancelling Headsets:** Uses an electronic system to cancel unwanted background noise while at the same time enhancing the quality of audio delivered through the headset.
- **Band-Type Hearing Protectors:** Designed for convenience in work areas with varying noise levels.
- **Communication Headsets:** Blocks unwanted noise while at the same time allowing the wearer to communicate clearly with coworkers. Special microphones suppress environmental noise to aid in two-way communication.
- **Ear Caps:** Seals the opening to the ear by entering the ear canal. Similar to band-type hearing protectors, they usually come on a band that be placed around the neck when the caps are in use for convenience in work areas with varying noise levels.
- **Ear Muffs:** Suppresses unwanted noise by completely covering the ear.
- **Disposable Ear Plugs:** Are made of formable material and are designed to be inserted into a person's ear canal.
- **Reusable Ear Plugs:** Are usually pre-molded and made from silicone, plastic, or rubber and are available in several different sizes. They are sometimes referred to as "Christmas tree plugs" because of their appearance.

NOTE: Headphones that are made for listening to music or other entertainment are NOT appropriate hearing protection. These types of headphones are not made to lower or block the noise of the environment or pieces of equipment.

SAFE WORK PRACTICES

When an employee works in an environment or with equipment that exceeds 90 decibels, they should wear the appropriate hearing protection. When choosing and using hearing protection, employees should do the following:

- Inspect all hearing protection for damage and a comfortable fit. Do NOT wear damaged or improperly fitting hearing protection. Report all damaged hearing protection to your supervisor.
- Ensure that the hearing protection chosen fits with the work that you will be doing.
- Wear all hearing protection in accordance with the manufacturer's instructions.
- Do NOT remove hearing protection until you have either turned off the piece of equipment being used or you have left the area.
- Do NOT use headphones that are meant for music or other entertainment in place of proper hearing protection.
- Store all hearing protection in accordance with the manufacturer's instructions.

CONCLUSION

NIHL is a condition that employees can develop over time if they do not wear the appropriate hearing protection while working in loud environments or with loud pieces of equipment. However, this condition is preventable. Employees need to remember that are responsible for their safety, including their hearing.

EXCAVATION: GUARDRAILS

INTRODUCTION

Guardrails are an essential part in fall protection for both the public and employees. Guardrails are required in most, if not all, construction projects, as they tend to be the number one method of preventing falls in the workplace. For the public, they act as a visible warning that an area is being worked on and that, for their safety, distance should be maintained. For employees it is a reminder to watch their activities around the excavation site and is an accident prevention tool. Following the safe work practices provided in this lesson will help ensure employee safety.

TYPES OF GUARDRAILS

Guardrails can be made from different materials, including:

- Wood
- Cable/Steel Cable
- Steel
- Aluminum

HAZARDS

Guardrails can pose some risks during an excavation, including:

- Damage to the guardrail. A damaged guardrail could cause the rail to fail if someone is leaning or looking over it. Or if the rail gets knocked over into the excavation, potentially hitting those in the excavation below. Damage usually occurs due to it being moved around the site, or when shoring is being installed.
- Having pipes or other materials lifted higher than needed, causing the boom (excavator, crane, etc.) to get too close to powerlines or other overhead obstructions
- Employees getting caught between railing and materials or excavator bucket
- Employees being encouraged to approach the excavation site due to the appearance of safety. (It gets employees closer to the edge of the excavation. Unless you are working in the excavation, you need to stay away from the edge at all times. Exceptions being the top person or competent person.)

PERSONAL PROTECTION EQUIPMENT

Employees should wear the following personal protective equipment:

- Hard hat
- Reflective vest or jacket
- Gloves
- Work safety boots

EXCAVATION: GUARDRAILS

SAFE WORK PRACTICES FOR EXCAVATION SITES

- Check the visibility of the excavation site; if it cannot be seen due to obstructions (trees, bushes, it's happening behind a building, etc.) then a guardrail system should be placed around the site.
- Inspect the guardrails for any rough or jagged surfaces. If the guardrail has such a surface, do not use it as it could cause punctures, lacerations, or snagged clothing.
- Double check the depth of the excavation site; if it is 6 feet or deeper, than a guardrail should be installed.
- All guardrails installed should have a top rail and a mid-rail.
- If you believe a hazard of falling objects exists, include a toe rail at the bottom of the guardrail system to prevent those working in the excavation site from being hit.
- Check the height of the guardrail. The top rail should be 42 inches plus or minus 3 inches.
- All guardrails should be able to withstand a force of at lest 200 pounds.
- If you are using a steel cable guardrail, check to ensure that it has flags every six feet. This is to help make the guardrail visible to others on the excavation site.
- A gate should be added to any guardrail that is used at a point of access (ladderways).
- Ensure that if you are lifting material over a guardrail, that it is not lifted any higher than necessary to clear the guardrail. Lifting material higher than needed could lead to the boom of the lift machine (crane, etc.) hitting a powerline or any other overhead obstruction.
- Should employees require a bridge/walkway to cross the excavation, guardrails should be installed.
- A bridge/walkway with guardrails should be installed at any excavation site where the bridge/walkway will be 6 feet or higher above the site.
- If using a steel cable guardrail system, the clamps should be placed correctly. That means that the U-blot and the Saddle are clamped with the U-blot facing down into the saddle which is facing straight up. The dead end of the cable should have small space between it and the body of the cable.
- When an excavation is not being worked on/in, there should be guardrails installed around the perimeter/edge of the site.

CONCLUSION

When safe work practices are followed, jobs can be completed in a safe and timely manner.

HEAT ILLNESS PREVENTION

INTRODUCTION

A healthy body temperature is maintained by the nervous system. As the body temperature increases, the body tries to maintain its normal temperature by transferring heat. Through sweating and blood flow to the skin, our bodies cool down. A heat-related illness occurs when our bodies can no longer transfer enough heat to keep us cool.
Four environmental factors affect the amount of stress a worker faces in a hot work area:

- Temperature
- Humidity
- Radiant heat (such as form the sun or a furnace)
- Air velocity

Other factors that can increase the possibility of heat-related illness are:

- Personal characteristics
- Age
- Weight
- Fitness
- Medical conditions
- Acclimation to heat

HEAT-RELATED ILLNESSES

When the body cannot dispose of excess heat, it will store it. Heat-related illnesses produce a high body temperature, which raises the body's core temperature and the heart rate increases. The following illnesses can occur when this happens:

- Heat rash (prickly heat), which occurs when the sweat ducts to the skin become blocked or swell. Symptoms are:
  - Scaly or red skin
  - Bumps or sores
  - Itchy or burning skin
- Heat cramps, which occur in muscles because sweating causes the body to lose water, salt, and minerals (electrolytes). Symptoms are:
  - Involuntary cramps
  - Intermittent cramps (they come and go)
- Heat edema (swelling), which can occur when you sit or stand for a long time in a hot environment. Symptoms are:
  - Swelling of the legs
  - Swelling of the ankles
  - Swelling of the hands
- Heat tetany (hyperventilation and heat stress), which is usually caused by short periods of stress in a hot environment. Symptoms are:
  - Hyperventilation
  - Respiratory problems
  - Numbness or tingling
  - Muscle Spasms
- Heat syncope (fainting), which occurs from low blood pressure when heat causes the blood vessels to expand (dilate) and body fluids move into the legs because of gravity. Symptoms are:
  - Dizziness
  - Dry Mouth
  - Excessive thirst
  - Headache

  - Nausea
  - Vomiting
  - Excessive sweating
  - Fainting
- Heat exhaustion (heat prostration), which generally develops when a person is working or exercising in hot weather and does not drink enough liquids to replace those lost liquids. Symptoms are:
  - The worker can still sweat
  - Weakness
  - Fatigue
  - Giddiness
  - Nausea
  - Headache
  - Clammy, moist skin
  - Redden or flushed complexion
  - Slightly raised body temperature
- Heatstroke (sunstroke), which occurs when the body fails to regulate its own temperature and body temperature continues to rise, often to 105°F (40.6°C) or higher. Symptoms are:
  - High body temperature
  - Absence of sweating
  - Rapid pulse
  - Difficulty breathing
  - Strange behavior
  - Hallucinations
  - Confusion
  - Agitation
  - Disorientation
  - Seizure
  - Coma

Someone showing these symptoms should be moved to a cool place, have their unnecessary garments removed and their body cooled. Lost fluids and electrolytes should be replaced. Consult a doctor if the person can't keep the fluid down or doesn't recover promptly. Heatstroke is a medical emergency. Even with immediate treatment, it can be life-threatening or cause serious long-term problems.

CONCLUSION

Most heat related health problems can be prevented or risk reduced by following a few basic procedures.

- Good ventilation of an indoor facility
- Fans, evaporative cooling or mechanical refrigeration
- Acclimatization using short exposures followed by longer periods of work in the hot environment.
- Drink plenty of water
- Take frequent shade breaks
- Stay away from caffeinated drinks when working in hot environments
- Learn to recognize the symptoms of heat-related illnesses
- Use protective equipment (Hats, cool fabrics, etc.)

EXCAVATION SAFETY: PREVENTING CAVE-INS

INTRODUCTION

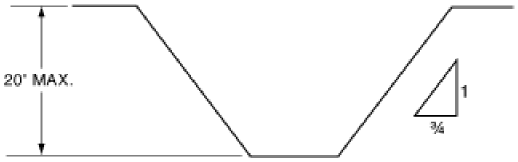
Excavating and trenching have many dangers, including the potential for cave-in. The type of digging you do will determine the safety precautions you will use.

CHOOSING PROTECTIVE SYSTEMS

- If the trench will be less than 5 feet deep, no protective system is required.
- If the excavation is entirely in stable rock, vertical sides may be used for protection against cave-ins.
- If there is any other type of excavation, it must be sloped, shored, or shielded.

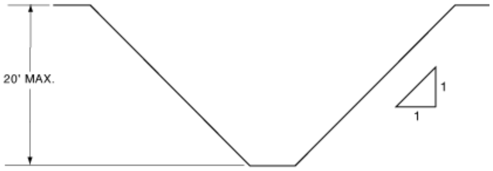
SLOPING/ BENCHING

Sloping is accomplished by cutting the banks of the excavation back to the angle of safety. At this angle the soil won't slide. This angle depends on the soil type.
Benching is a method of protecting employees from cave-ins by excavating the sides of a trench to form one or a series of horizontal levels or steps, usually with vertical or near-vertical surfaces between levels. The ratio of the horizontal and vertical cuts also depends on the soil type.



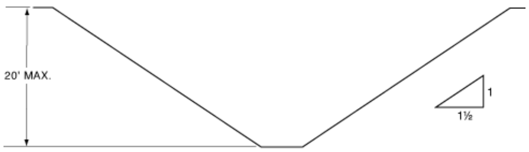
SIMPLE SLOPE - GENERAL

- Type A: Type A soil is solid or made up of clay
  - The slope angle and bench ratio must be no less than ¾ horizontal for every 1 vertical.



SIMPLE SLOPE

- Type B: Type B is unstable dry rock.
  - The slope angle and bench ratio must be 1 horizontal for every 1 vertical.



SIMPLE SLOPE

- Type C: Type C is granular soil (like gravel or sand), submerged soil, or soil from which water is freely seeping.
  - The slope angle must be 1 ½ horizontal for every 1 vertical (There is no bench ratio because benching cannot usually be done in Type C soil).

These requirements are only good for trenches that are 20 feet deep or less; any deeper and there must be a different system.

EXCAVATION SAFETY: PREVENTING CAVE-INS

SHORING/ SHIELDS

Shoring is a structure such as a metal hydraulic, mechanical, or timber bracing system that supports the sides of an excavation. A shield (also known as a trench box) is a structure that is able to withstand the forces imposed on it by a cave-in and thereby protect employees within the structure. Shields can be permanent structures or can be designed to be portable and moved along as work progresses.

ACCESS/EGRESS

No matter what protective system is required, you must always have the proper access and egress for the trench:

- A stairway, ladder, ramp, or other safe means of access/egress shall be located in trench excavations that are 4 feet or more in depth.
- The means of access/egress shall not be more than 25 feet away from where the worker is working.

GENERAL SAFETY

Inspect the protective system daily to be sure it is still safe. Be aware of your surroundings and recognize unsafe conditions:

- Vibrations from equipment could endanger the shoring system.
- Pools of liquid, blowing dirt, hissing sounds, vapor clouds, and/or gaseous odors may mean a utility pipe or wire was damaged.
- Wet or discolored soil is an indication of a water/sewer leak and should be treated as a potential emergency condition.

CONCLUSION

You can stay safe from cave-ins if you:

- Use the correct protective systems.
- Have a safe means of access/egress.
- Stay aware of your surroundings and recognize unsafe conditions.

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HEAVY EQUIPMENT SAFETY

INTRODUCTION

Heavy equipment such as tractors, excavators, dump trucks, cranes, forklifts, etc., are common sights on any construction site. Heavy equipment is used for a variety of tasks, and both operators and those who work around them could potentially be exposed to hazards. Following the safe work practices presented in this lesson will help ensure employee safety.

HAZARDS

When operating or working around heavy equipment, employees could potentially be exposed to the following hazards:

- Being run over by equipment
- Being caught in/or between equipment
- Falls
- Electrocution (when equipment makes contact with power lines or when performing maintenance tasks)
- Rollovers
- Being crushed by falling elevated components
- Hearing loss

PERSONAL PROTECTIVE EQUIPMENT (PPE)

Employees should wear the appropriate personal protective equipment, which may include:

- Hi-vis Reflective vest, jacket, and/or clothing
- Hard hat
- Safety glasses
- Gloves
- Steel-toed shoes or boots
- Hearing protection

HEAVY EQUIPMENT SAFETY

SAFE WORK PRACTICES

Prior to operating a piece of heavy equipment, employees should do the following:

- Read the operator’s manual.
- Inspect the piece of equipment in accordance with the manufacturer’s instructions. Report pieces of equipment that have been modified, have worn parts, or are damaged. Do NOT operate modified or damaged pieces of equipment.
- Ensure that your piece of equipment has a fire extinguisher. Fire extinguisher should be inspected in accordance with all local, state, and federal laws. Report missing fire extinguishers to your supervisor.
- If the task requires a spotter, pre-plan how communication will work, such as if signals or walkie-talkies are going to be used. Ensure that both parties agree on the signals prior to starting operations.
- Know where your spotter is going to be located (if applicable).
- Communicate with others that will be working in the area about the blind spots of your piece of equipment. In addition, you should inform everyone of the swing radius of your piece of equipment if it has it.

When operating or working around heavy equipment, employees should do the following:

- Only authorized and trained employees should operate heavy equipment.
- Use your seatbelt.
- Maintain three-points of contact when entering or exiting your piece of equipment.
- Be aware of your surroundings.
- Operate the piece of equipment in accordance with the manufacturer’s instructions.
- Use alarms to alert others in the area when backing up.
- Use manufacturer-provided cameras to see behind and around you (if applicable).
- Stay in proper communication with the spotter (if applicable).
- For non-operator employees, before walking in front of a piece of equipment, make eye contact with the operator and wait for the piece of the equipment to come to a complete stop before walking.
- If needed, get out of the piece of equipment and do a walk around to ensure that no one is in your path of movement before moving your piece of equipment.
- Maintain designated distances from powerlines.
- When performing maintenance tasks, ensure all required lockout/tagout procedures are followed.

When operating or working around piece of heavy equipment, employees should NOT do the following:

- Do NOT walk behind any pieces of equipment.
- Do NOT leave loads in elevated positions.
- Do NOT walk under elevated loads.
- Do NOT jump down from pieces of equipment.
- Do NOT modify any pieces of equipment.
- Do NOT operate a piece of equipment without wearing your seatbelt.

CONCLUSION

While a common sight on construction sites, heavy equipment can potentially expose operators and non-operator employees to harm. By following the safe work practices presented in this lesson, employees can help minimize their chances of an accident occurring when operating or working around pieces of heavy equipment.

FALL PROTECTION: CONSTRUCTION

INTRODUCTION

In construction, falls account for one of the leading causes of injuries and fatalities for employees. To help minimize the chances of injuries or death, employees should use the appropriate method of fall protection for their task. Following the safe work practices presented in this lesson will help ensure employee safety.

WHEN IS FALL PROTECTION REQUIRED?

Under OSHA, employees should use fall protection when performing tasks on walking or working surfaces that have unprotected sides or edges that are 6 feet or more above a lower level. Examples include, but are not limited to:

- Wall openings
- Holes
- Excavations
- Roofs
- Hoist areas
- Leading edges

Note: In California, fall protection is required at 7 ½ feet.

HAZARDS

When working at heights, employees could potentially be exposed to some of the following injuries should a fall occur:

- Head injuries (including fractures to the skull, traumatic brain injuries, etc.)
- Broken bones (including wrist, arm, ankle, and hip fractures)
- Impalement
- Death

TYPES OF FALL PROTECTION

There are different types of fall protection, including:

- Guardrails
- Personal fall arrest systems
- Safety nets
- Warning lines
- Controlled access zones
- Covers
- Fall Protection Plan
  - *When traditional fall protection creates a greater hazard, a specifically written plan that details why conventional fall protection creates a greater hazard or is infeasible may be used. The plan discusses other measures that will be taken to reduce or eliminate the fall hazard. This plan has to be written by a qualified person and developed specifically for the site where the hazard is located.*
  - *A safety monitoring system is part of a fall protection plan. This system requires that a designated competent person monitors the area where a fall protection plan has been implemented. His or her job is to warn other employees about the fall hazard by keeping unauthorized employees out of the area, stay in visual sighting distance of employees being monitored, and have no other responsibilities which could take their attention away from their monitoring duties.*

FALL PROTECTION: CONSTRUCTION

SAFE WORK PRACTICES

Before working in an area that requires fall protection, employees should do the following:

- Only trained and authorized employees should work in areas that require fall protection.
- Ensure that you are wearing the appropriate personal protective equipment for the area and task. PPE may include hard hats, gloves, safety glasses or goggles, steel-toed shoes, etc.
- Inspect provided fall harnesses, including lanyards, for wear or damage (if applicable). Report worn or damaged harnesses to your supervisor. Do NOT wear a worn or damaged fall harness.
- Become familiar with your company's fall protection plan (if applicable). If you have any questions about the plan, ask your supervisor for clarification.

When working at heights requiring fall protection, employees should do the following:

- Follow your company's fall protection policies.
- Use the appropriate fall protection method for the area and task.
- Ensure that all guardrails are erected in accordance with all manufacturer's instructions and OSHA regulations (if applicable). Guardrails should be inspected for signs of damage before being set up and while they are in use. Report damaged guardrails to your supervisor. Do NOT set up damaged guardrails.
- Ensure all personal fall arrest systems are set up and used in accordance with the manufacturer's instructions and OSHA regulations (if applicable).
- Ensure that all safety nets are constructed in accordance with the manufacturer's instructions and OSHA regulations (if applicable). Drop-testing of safety nets should be done according to OSHA regulations.
- Ensure that all warning lines are set up in accordance with OSHA regulations and are flagged with high-visibility material.
- Ensure that all controlled access zones have the appropriate signage to warn unauthorized employees against entry.
- Appropriately cover and secure all holes in accordance with OSHA regulations.
- Only qualified and trained employees should perform work in controlled access zones.
- In areas where there is the potential for falling objects, either use a canopy structure or erect barricades to protect workers on lower levels. An example of a barricade would be a toe board on a guardrail.
- Ensure all items that could become falling objects are properly stored when not in use.
- Immediately report a fall to your supervisor. Follow your company's rescue plan in the event of a fall occurring.

When working at heights requiring fall protection, employees should NOT do the following:

- Do NOT attach personal fall arrest systems to guardrails.
- Do NOT enter a controlled-access zone unless you are authorized to do so.
- Do NOT work on edges unless the appropriate fall protection is in place.
- Do NOT modify or bypass any provided fall protection. Modified fall protection should be immediately reported to your supervisor.

CONCLUSION

Falling is one of the most dangerous hazards construction workers can encounter while on the jobsite. Falls can result in injuries such as concussions, broken or fractured bones, impalement, etc. By following the safe work practices presented in this lesson, employees can help minimize their chances of a fall occurring while working at dangerous heights.

EXCAVATION SAFETY: UTILITIES

INTRODUCTION

In order to save space and keep the area looking nice, many utility lines, pipes, and cables are put underground. However, these unseen utilities present a hazard for excavation projects. Therefore, follow these tips:

811/UNDERGROUND SERVICE ALERT

The Underground Service Alert is a program set up specifically to prevent utility accidents during excavation. Excavators are to call 811 at least two working days before the planned excavation. Underground Service Alert will then contact the utility companies, who will come and mark the location of their utilities within the planned excavation site. The five steps to a safe excavation are:

1. Survey and mark the site
2. Call/Inform online
3. Wait the required time
4. Respect the Marks
5. Dig with Care

Hopefully following the steps for the underground service alert will prevent any accidents with utilities. However, if an accident does happen, always:

- Contact the utility operator
- Call 911 or the local fire/police department to alert them to any potential hazards

ELECTRICITY

If the excavation comes into contact with overhead or electric lines:

- Turn off the equipment ONLY if you can safely do so.
- If you are on the equipment that is in contact with the electric utilities, either stay on the equipment or jump clear of it.
    - You will get electrocuted if you touch the equipment and the ground at the same time.

EXCAVATION SAFETY: UTILITIES

WATER/SEWER

If the excavation causes water or sewage lines to break:

- Evacuate immediately- trenches can fill up quickly.
- Do not close valves to prevent flooding.
- Be careful of high pressure water lines because even the slightest scratch or vibration can cause the lines to burst.
- Move carefully around trenches with wet walls.
- Avoid contact with waste water.
- Sewer gas is flammable- avoid open flames or anything that might start a fire.

FIBER OPTIC CABLES/TELECOMMUNICATIONS

- If you cut into a fiber-optic cable, do not look directly into it.

CONCLUSION

Hopefully excavators will avoid contact with utilities if they dial 811/contact the Underground Service Alert in their area. However, if something does happen, knowing what to do will help prevent any serious consequences during excavation.

EXCAVATOR SAFE OPERATION

INTRODUCTION

Excavators are a type of earth moving equipment that can move a massive amount of dirt in a short period of time. They can also cause serious injury if not handled correctly, so it is important to follow all safety guidelines while operating one.

GENERAL SAFETY RULES

The operator has full responsibility of the excavator at all times. To reduce the chance of an injury either to yourself or others, the following safe work practices are recommended:

- Buckle the seatbelt before starting the engine.
- Scan the area to make sure that no one is in the intended path.
- Double check that the control pattern is set to your desired mode and that all levers are performing their indicated task.
- Select a route that is flat as possible when driving, and slow when going over rough terrain.
- To clean the bucket, do not strike it against the ground or another object.
- Make small and gradual changes to turn.
- Never allow riders anywhere on the excavator or bucket.
- Do not operate the excavator from anywhere other than the seat.
- Carry the bucket low to increase stability and visibility.
- Inspect the jobsite if you are unfamiliar with it to become aware of any slopes, banks, or overhead obstacles (such as power lines).
- Park the machine on a level surface with the bucket or attachment and blade lowered to the ground. When exiting, remember to face the machine and maintain three points of contact on the hand or foot holds at all times.

TRENCHING OR DIGGING

- When trenching, make sure the excavator is level.
- Spoil piles must be dumped a sufficient distance away to prevent cave-ins.
- Do not dig near the edge of an excavation or trench
- Never dig underneath the excavator.
- Underground utilities should always be marked before digging.

CLIMBING OR DESCENDING HILLS

Hills are not ideal conditions to operate an excavator, but in the event that you are required to do this, you should know how to do so safely.

- Do not drive diagonally or horizontally across slopes. When driving up a slope, the boom and arm should be extended with the bucket carried low. If the unit begins to slide, you can set the bucket down to prevent sliding.
- In very steep conditions, you can use the boom and arm to assist when moving up or down the slope. When going up, alternate extending the arm and retracting it towards you to help lift the excavator up the slope.
- When going down a hill or embankment, position the bucket with the flat surface resting on the ground to support the unit and reduce the chance of slipping.

CONCLUSION

Whether you are trenching, climbing hills, or just maneuvering around the jobsite, all aspects of excavator handling can be made safer if the included safety rules are followed.

EXCAVATIONS AND TRENCHING

INTRODUCTION

Excavating and trenching work are the starting point of most construction projects. There are many considerations that must be made while trenching or excavating.

MAJOR CAUSES OF CAVE-INS

- Inadequate shoring
- Poor analysis of soil conditions
- Defective shoring materials
- Nearby loads
- Vibrations
- Weather conditions

ELIMINATE EXCAVATION HAZARDS

- Check and locate any underground utilities or other buried items. “Call Before You Dig.”
- Soil conditions must be carefully evaluated to determine the protective system needed.
- Wear your hard hat at all times as well as rugged boots.
- Excavate trenching banks to their proper slope ratio.
- Where necessary, straight banks should be shored.
- Weather conditions can greatly affect sloping and shoring.
- Material stockpiled nearby can increase the pressure on trench or excavation walls.
- Keep heavy equipment and materials such as pipe and timbers well away from the excavation site.
- Maintain a minimum of two feet between any materials, including the spoils pile and the edge of the trench.
- Vibrations from equipment passing by can contribute to cave-ins by loosening the soil. Any soil vibration can endanger a shoring system.
- Compaction operations cause vibration; therefore, check soil conditions before, during and after compaction.
- A competent person is to inspect shoring systems daily.
- Ladders are to be located no more than 25 feet away from any worker. Ladders must extend from the floor of the excavation to 3 feet above the top and must be secured at the top.

PLAN BEFORE YOU START EXCAVATING

- Inspect the site.
- Collect information.
- List the risks.
- Mitigate or eliminate potential problems
- Establish minimum rate of inspection

- Have written site safety plan, including emergency procedures

OSHA COMPETENT TRENCHING AND EXCAVATION PERSON

OSHA requires that a competent person manage all excavation work. That person has to:
- Be on the job site each and every time someone is working in an excavation area.
- Have knowledge of OSHA soil-types and know how to make both a visual and a physical inspection of the soil.
- Have knowledge about the protective systems or methods used to keep workers safe.
- Know how to inspect each excavation before the start of each workday as well as after any change of conditions in the trench or excavation.
- Know when to take action directing workers to get out of dangerous trenches or excavations.
- Understand protection choices such as:
    - Benching
    - Shoring
    - Terms
    - Selection
    - Installation and removal
    - Effect of water

CONCLUSION

Plan before you start any excavations or trenching with a competent person managing all the work. Follow all OSHA requirements and make sure your trenches and excavations are safe.

MUSCULOSKELETAL INJURY PREVENTION (CONSTRUCTION)

INTRODUCTION

One of the leading causes of non-fatal injuries in construction are musculoskeletal disorders (MSDs). MSDs may also be referred to as work-related musculoskeletal disorders (WMSDs), as many MSDs are related to tasks performed in the work environment. Many MSDs are attributed to the repetitive work that construction workers routinely perform while on the jobsite. MSDs are preventable injuries, but for those who are diagnosed with an MSD, the effects can lead to long term recovery or be permanent. Following the safe work practices presented in this lesson will help ensure employee safety.

WHAT ACTIVITES ARE THE RISKIEST FOR DEVELOPING MSDs IN THE CONSTRUCTION INDUSTRY?

In the construction industry, many MSDs develop over long periods of time caused by the tasks employees are repeatedly exposed to or by performing repetitive motions. Such tasks include:

- Continuous exposure to vibrations (jackhammers, power tools, etc.)
- Performing multiple manual lifts
- Lifting items or materials that are too heavy
- Continuously maintaining awkward postures
- Repetitive motions such as hammering, reaching, bending, etc.
- Pushing, pulling, tugging, twisting, and/or sliding

MUSCULOSKELETAL INJURY PREVENTION (CONSTRUCTION)

SAFE WORK PRACTICES

Housekeeping

To help in the reduction of MSDs, employees should ensure that the jobsite is clear of debris and obstructions as to avoid slipping, tripping, or falling. In addition to keeping the floors clear, work areas should also be well lit, and holes or openings properly covered.

Ergonomics

Employees should use proper ergonomic lifting techniques when performing manual lifts. Proper lifting technique includes:

- Bending your knees.
- Keeping your feet shoulder width apart.
- Keeping your back straight.
- Lifting close to your body.

Employees should NOT do the following when performing a manual lift:

- Do NOT lift materials or objects that are too heavy or awkward. These types of materials should be lifted by either using a team lift or using a mechanical lift assist.
- Do NOT twist the body when lifting. Employees should turn their body in the direction that they want to go, by turning their feet and completing a full turn.
- Do NOT overreach. Employees should use stools or step ladders to reach items that are too high and use tools that allow them to extend their reach comfortably.

Reduce Repetitive Motions

Employees should reduce the amount of repetitive motions that they perform throughout their shift. Employees can reduce repetitive motions by:

- Rotating tasks with other employees.
- Taking breaks and performing other tasks before returning to the repetitive task.
    - When operating equipment that exposes employees to vibrations, it is recommended that employees switch at designated times to reduce exposure.
- Using a mechanical option when it is available.
- Maintaining good posture while performing repetitive tasks.

Tools and Equipment

When it's possible, employees should use tools and equipment that will help in the reduction of tasks that contribute to MSDs. This includes using tools such as extension poles, extension handles, kneeling creepers, auto-feed screw guns, rebar tying tools, etc.

CONCLUSION

Musculoskeletal disorders in the construction industry are typically caused by repeated exposure to certain tasks or repeating certain movements throughout the day. Employees can help reduce their chances of developing MSDs by practicing proper ergonomics and reducing the amount of repetitive motions that they perform throughout their shift. Keeping the jobsite clean is another way employees can help in the reduction of MSDs.